Manufacturing

"OUR PATENTED METHOD Now Extracts Lithium Carbonate At Over 90% EFFICIENCY And 99% PURITY —Up From An Initial 35%!"

In a conversation with EFY's Nitisha Dubey, Prassann Daphal of Evergreen Recyclekaro India Private Limited highlights how plasma furnace technology transforms complex materials into a molten composite, primed for advanced hydrometallurgical extraction.

PRASSANN DAPHAL
CEO, EVERGREEN RECYCLEKARO INDIA PRIVATE LIMITED

How does Recyclekaro recover valuable metals from electronic waste (e-waste) and lithium-ion batteries?

When we recycle an item like a laptop, the first step is dismantling the unit and carefully separating its base materials. We remove the motherboard and strip the components, categorising each based on its material content. For instance, the integrated circuits (ICs), rich in precious metals, are shredded into powder. This powder is then subjected to advanced metallurgical processes, such as solvent extraction, to recover valuable metals like gold, silver, platinum, and palladium. When handling lithium-ion batteries, we ensure that critical metals such as lithium carbonate, cobalt sulphate, magnesium sulphate, nickel sulphate, and graphite are extracted. The laptop casing, often made from aluminium, plastic, or iron, is separated and processed accordingly.

What types of machines and equipment does Recyclekaro use?

We operate a manufacturing facility in Vada, Palghar, with a capacity of 4200 metric tonnes per